HOW FAR JOHANNES KIRMSE GOT IN 1924: A VERIFIED EXPOSITION OF HIS ARITHMETIC OF THE INTEGRAL OCTONIONS

CLAUDE OPUS 4.7

Abstract. This is a self-contained exposition of Johannes Kirmse's 1924 paper on the arithmetic of the octonions [1], written by the AI agent at the direction of Jens Köplinger as part of the research record of the repository [2]; the verification it reports was carried out under prompts 100 and 103 of that record. Its single purpose is to establish, from Kirmse's original text and by independent computation, *exactly* what Kirmse achieved and where his one substantive error lies.

The conclusion is this. Kirmse built the arithmetic theory of the integral octonions in full: the units and the multiplication table, conjugation and norm, the composition identities, modules, orders and maximal orders, an ideal theory, and from it the classical eight-squares theorem. His one substantive error is a miscount: he states ([1], p. 70) that there are *eight* maximal orders containing the integer lattice J_0, whereas there are *seven*; and the one maximal order he writes down explicitly, his J_1, is itself *not* closed under multiplication. Every formula below is traced to Kirmse's original; each section closes with a *Sources and tracing* remark.

The companion historical questions (who corrected the count, in what form, and where the error first entered the later literature) depend on Coxeter's 1946 paper, which is re-derived in exact arithmetic in this project (`python_project/src/verify_coxeter_1946.py`; verification note in `paper/reviews/`, 22 May 2026). This exposition is kept as the internal Kirmse-only record; the public-facing condensation that synthesizes Dickson, Kirmse, Mahler, and Coxeter is Appendix B of the main paper (drafts: `paper/historical_appendix.tex` and its expanded form `paper/historical_appendix_expanded.tex`). An appendix records, as explicit speculation and at the suggestion of J. Köplinger, one possible human provenance of the error.

1. Purpose and method

Johannes Kirmse's name survives in two phrases. To workers in nonassociative algebra it survives in *Kirmse's identities*; to those concerned with integral quadratic forms it survives in *Kirmse's mistake*. The second phrase has all but erased the first, and with it the fact that Kirmse's short 1924 paper is one of the founding documents of the arithmetic of the octonions.

The present note was written to answer one question precisely: how far did Kirmse get, and where exactly is the mistake? It follows the methodology of the repository [2] in which it was produced: no claim (Kirmse's or anyone's) is taken on trust; every assertion is either traced to Kirmse's printed text or verified from scratch by exact computation. The two are kept strictly separate. Sections 2–4 expound Kirmse's paper and trace every formula to it. Section 5 reports an independent computation, owing nothing to Kirmse or to any later author, that tests his central claim. Section 6 draws the balance.

We write $\mathbb{O}$ for the real algebra Kirmse studies (the Graves–Cayley octonions) and reserve the symbols J_0, J_1 for the two lattices he names.

2. The elementary arithmetic of the Cayley numbers

Kirmse begins with an algebra of *eight* units $i_0, i_1, \ldots, i_7$, with $i_0 = 1$ the identity, and a real algebra

$$a = \alpha_0 + \alpha_1 i_1 + \alpha_2 i_2 + \cdots + \alpha_7 i_7, \qquad \alpha_0, \ldots, \alpha_7 \in \mathbb{R}.$$

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The multiplication of the units is fixed by his table (1). Reading that table, the seven products that carry a positive sign organize into seven oriented triples

$$(1,2,3),\ (1,5,4),\ (1,6,7),\ (2,4,7),\ (2,6,5),\ (3,4,6),\ (3,5,7), \tag{2.1}$$

with the rule: for each triple (a,b,c),

$$i_a i_b = i_c, \quad i_b i_c = i_a, \quad i_c i_a = i_b \qquad \text{(and the reverse products carry a minus sign)},$$

together with $i_j^2 = -1$ for $j = 1,\dots,7$ and $i_0 i_j = i_j i_0 = i_j$. Written out, the products of the seven imaginary units are

$\cdot$	i_1	i_2	i_3	i_4	i_5	i_6	i_7
i_1	-1	i_3	$-i_2$	$-i_5$	i_4	i_7	$-i_6$
i_2	$-i_3$	-1	i_1	i_7	$-i_6$	i_5	$-i_4$
i_3	i_2	$-i_1$	-1	i_6	i_7	$-i_4$	$-i_5$
i_4	i_5	$-i_7$	$-i_6$	-1	$-i_1$	i_3	i_2
i_5	$-i_4$	i_6	$-i_7$	i_1	-1	$-i_2$	i_3
i_6	$-i_7$	$-i_5$	i_4	$-i_3$	i_2	-1	i_1
i_7	i_6	i_4	i_5	$-i_2$	$-i_3$	$-i_1$	-1

This is Kirmse's table (1) in full. He notes that the problem he set himself (an eight-unit system containing the quaternions and carrying the fourth power of the sum of squares as a norm) admits "several, but not essentially different" solutions, of which (1) is one.

To each a Kirmse associates the *symmetric* number (the conjugate) and the *Relativnorm*

$$\overline{a} = \alpha_0 - \alpha_1 i_1 - \cdots - \alpha_7 i_7, \tag{2.2}$$

$$\mathrm{N}(a) = a\overline{a} = \sum_{j=0}^{7} \alpha_j^2. \tag{2.3}$$

Conjugation reverses products,

$$\overline{ab} = \overline{b}\,\overline{a}. \tag{2.4}$$

Developing the eight products $a, ai_1, \dots, ai_7$ in the units yields an 8×8 matrix $M[a]$ of a linear substitution; its determinant, the *Grundnorm*, equals $\mathrm{N}(a)^4$, and from the structure of $M[ab]$ Kirmse obtains the composition law

$$\mathrm{N}(ab) = \mathrm{N}(a)\,\mathrm{N}(b). \tag{2.5}$$

Calling a *regular* when $\mathrm{N}(a) \neq 0$, he solves the one-sided equations $ax = c$ and $ya = c$:

$$x = \frac{\overline{a}\,c}{\mathrm{N}(a)}, \qquad y = \frac{c\,\overline{a}}{\mathrm{N}(a)}. \tag{2.6}$$

Specializing $c = ab$ gives the two *composition identities*

$$\overline{a}\,(ab) = \mathrm{N}(a)\,b, \qquad (ab)\,\overline{b} = \mathrm{N}(b)\,a, \tag{2.7}$$

so that "for products of three factors in which two consecutive factors are conjugate, the associative law holds": in particular $(\overline{a}a)b = \overline{a}(ab)$ and $a(b\overline{b}) = (ab)\overline{b}$. Finally Kirmse calls a_1, a_2, a_3 *coplanar* when corresponding imaginary components are proportional; coplanar numbers commute and associate, every power a^m is coplanar with a, and "coplanar numbers behave like algebraic numbers."

Sources and tracing. The eight units and the multiplication table reproduce Kirmse's table (1), [1] p. 64; we have rendered his printed array of 64 products $i_j i_k$ as the seven oriented triples (2.1) plus $i_j^2 = -1$ and $i_0 i_k = i_k$, and the 7×7 array above is that same table written out (it was checked entry-by-entry against the scan of p. 64). The conjugate (2.2) and norm (2.3) are Kirmse's, p. 64 (he names $\overline{a}$ the number "symmetric to a" and $\mathrm{N}(a)$ the *Relativnorm*). Equation (2.4) is Kirmse's (2), p. 64. The matrix $M[a]$ is his (3) and the product expansion his (4), p. 65; the composition law (2.5) is his (5), p. 66, derived there from the column structure of $M[ab]$. The division formulas (2.6) are his (7 a) and (7 b), pp. 66–67; the composition identities (2.7) are the content of his (8), p. 67. The remarks on coplanar numbers follow his (9)–(10), p. 67. All of this is Kirmse's §1, "Die elementare Arithmetik der Cayleyschen Zahlen."

3. Modules, orders, and the count of maximal orders

Kirmse now restricts to the octonions with rational components, K (those a with all $\alpha_j \in \mathbb{Q}$), and develops a theory of lattices in K.

Definition 3.1. A *module* is a subset of K closed under integer linear combinations. It is *finite* if it consists of all $\gamma_0 a_0 + \cdots + \gamma_7 a_7$ ($\gamma_j \in \mathbb{Z}$) for eight independent numbers $a_0, \dots, a_7$. A finite module is an *Integritätsbereich* (an *order*) if with any two numbers a, b it also contains the products ab and ba. An order is a *most-comprehensive order* (*umfassendster Integritätsbereich*), that is, a *maximal order*, if it contains

$$J_0 = [\,1, i_1, i_2, i_3, i_4, i_5, i_6, i_7\,] \tag{3.1}$$

and cannot be enlarged by adjoining a further number of K without ceasing to be an order.

The lattice J_0 of octonions with integer components is itself an order. Kirmse's structural theorem about *any* order is obtained as follows. If a lies in an order with $\mathbb{Z}$-basis $a_0, \dots, a_7$, the products aa_j have integer coordinates in that basis; eliminating the basis from the eight resulting linear relations gives an integer determinant $D[a]$ with

$$D[a] = 0, \tag{3.2}$$

so every number of an order satisfies a monic integral equation of degree 8. Because conjugates and coplanar combinations again satisfy such equations, $a + \overline{a}$ and $a\overline{a}$ are rational integers; and multiplying by a suitable unit moves any chosen component to the front. Kirmse concludes:

Theorem 3.2 (Kirmse, p. 69)**.** *The double of every number belonging to an order $J \subseteq K$ has exclusively integer components.*

Thus the numbers of an order have components that are integers or half-integers, and the maximal orders may be sought by a finite search among lattices between J_0 and $\frac{1}{2}J_0$. At this point Kirmse writes ([1], p. 70), verbatim:

> *"Dieser Satz erlaubt sofort, durch eine einfache endliche Rechnung die sämtlichen umfassendsten Integritätsbereiche über J_0 aufzusuchen.* **Man findet acht solche***, deren jeder die Gestalt hat"*

$$J = \Big[\, J_0,\ \tfrac{1}{2}\big(i_{\nu_1} + i_{\nu_2} + i_{\nu_3} + i_{\nu_4}\big),\ \tfrac{1}{2}\textstyle\sum_{k=0}^{7} i_k \,\Big], \tag{3.3}$$

the indices $\nu_1, \nu_2, \nu_3, \nu_4$ all distinct, drawn from $\{0, 1, \dots, 7\}$. That is: *Kirmse asserts there are eight maximal orders.* He adds that each contains exactly fourteen elements of the "middle kind" $\frac{1}{2}$(four units), falling into seven complementary pairs.

He then exhibits *one* of the eight explicitly. Calling it J_1,

$$J_1 = \big[\, J_0,\ a_1,\ a_2,\ a_3,\ a_4 \,\big], \tag{3.4}$$

$$a_1 = \tfrac{1}{2}\big(1 + i_1 + i_2 + i_3\big), \quad a_2 = \tfrac{1}{2}\big(i_4 + i_5 + i_6 + i_7\big),$$
$$a_3 = \tfrac{1}{2}\big(1 + i_1 + i_4 + i_5\big), \quad a_4 = \tfrac{1}{2}\big(1 + i_2 + i_4 + i_7\big), \tag{3.5}$$

with J_0 of index $\mathrm{N}(J_1) = 16$ in J_1. Every later section of the paper works in this J_1.

Sources and tracing. The definitions of module, finite module, order (*Integritätsbereich*) and maximal order (*umfassendster Integritätsbereich*), and the lattice J_0 of (3.1), are Kirmse's, [1] p. 68. The integral degree-8 equation (3.2) is his (11), p. 69, derived in his §2 by the elimination just summarized. Theorem 3.2 is the italicized *Satz* at the foot of his p. 69. The claim of eight maximal orders and the schematic form (3.3) are on his p. 70; the quoted German is verbatim, and the emphasis on "acht" (*eight*) is added. The order J_1 and its generators (3.4)–(3.5), and the index 16, are Kirmse's, p. 70. All of this is his §2, "Moduln und Integritätsbereiche im Gebiete der Cayleyschen Zahlen mit rationalen Komponenten."

4. What Kirmse built on J_1

The remaining three sections, all set in J_1, show the reach of the construction; we summarize them only to record how far Kirmse carried the theory.

In §3 ("Die ungeraden Ideale…") he develops one- and two-sided ideals, their norms, and primitive and primary ideals. In §4 ("Die Einheiten und die geraden Ideale in J_1") he finds the units: *sixteen* in J_0 (the $\pm i_j$) and *two hundred and forty* in J_1 (the $\pm i_j$ together with 224 half-integer units), and observes that they "form a group in an extended sense, in which all group-theoretic theorems hold except the associative law and its consequences," i.e. a Moufang loop; he counts $2160 = 240 \cdot 9$ numbers of norm 2. In §5 ("Die Primzerlegung…") he gives the prime decomposition, finds that every ideal of J_1 (and every odd ideal of J_0) is principal, computes the ideal zeta function

$$Z(s) = \zeta(4s)\,\zeta(4s-3), \tag{4.1}$$

and derives the eight-squares theorem: the number of representations of an odd $n > 0$ as a sum of eight integer squares is

$$r_8(n) = 16 \sum_{d \mid n} d^3. \tag{4.2}$$

A postscript adds a Minkowski geometry-of-numbers proof that all ideals are principal.

These results (the 240 units, (4.1), (4.2)) are correct. They are facts about the E_8 lattice as a quadratic form and, as Section 5 will make clear, do not depend on the one claim that fails.

Sources and tracing. The section titles and contents are Kirmse's §§3–5, [1] pp. 70–82: the ideal theory of §3 (from p. 70); the units and the count 240, the "group in an extended sense," and $2160 = 240{\cdot}9$ in §4, p. 76; the zeta function (4.1) (his $Z(s)$, p. 77) and the eight-squares theorem (4.2) (his numbered conclusions 2–3, p. 78); the Minkowski argument is the *Nachschrift*, p. 82. Equations (4.1)–(4.2) restate his results in his own notation.

5. Independent verification

Kirmse asserts the eight maximal orders of (3.3) "durch eine einfache endliche Rechnung," but prints no part of that computation. We carry it out, in exact arithmetic, from his data alone: the multiplication table (1) of Section 2 and the order J_1 of (3.4)–(3.5). Nothing from any later author enters.

(a) The table is sound. Built from the seven triples (2.1), the multiplication satisfies the composition law (2.5), $\mathrm{N}(uv) = \mathrm{N}(u)\mathrm{N}(v)$, on every tested pair. Kirmse's $\mathbb{O}$ is a genuine (composition) octonion algebra.

(b) J_1 is not closed. J_1 is the $\mathbb{Z}$-span of the twelve numbers $1, i_1, \ldots, i_7, a_1, a_2, a_3, a_4$. By bilinearity, J_1 is an order if and only if all 144 products of these generators lie in J_1. Ten of them do not. The simplest witness uses two of Kirmse's own generators:

$$a_1 \cdot a_3 = \tfrac{1}{2}(1+i_1+i_2+i_3) \cdot \tfrac{1}{2}(1+i_1+i_4+i_5) = \tfrac{1}{2}\big(i_1 + i_2 + i_4 + i_7\big). \tag{5.1}$$

The number $\frac{1}{2}(i_1 + i_2 + i_4 + i_7)$ does not belong to J_1: its half-integer pattern is not one of the sixteen that J_1 admits. Hence

$$J_1 \cdot J_1 \not\subseteq J_1;$$

J_1 is not closed under multiplication, and so is not an order at all.

(c) There are seven maximal orders, not eight. A maximal order containing J_0 is a unimodular even lattice, hence corresponds to a doubly-even self-dual binary code of length 8; there are exactly 30 such codes (the coordinate relabelings of the extended Hamming code). Testing each of the 30 candidate lattices for closure under table (1) leaves exactly

$$\boxed{\text{seven}}$$

that are closed. Kirmse's J_1 is among the 30 candidates but is one of the 23 that fail. His count (3.3) is therefore too large by one, and the representative he chose was one of the nonorders.

(d) The structure of the error. Closure of a candidate lattice is the conjunction of two independent sub-conditions: that every product of a unit with a half-integer element again lies in the lattice, and that every product of two half-integer elements does. The first sub-condition alone (invariance under multiplication by the units) is satisfied by exactly *eight* of the thirty candidates; imposing the second removes exactly one. The eight unit-invariant lattices are precisely the seven genuine maximal orders together with Kirmse's J_1, and J_1 is the unique one of the eight that is not an order. Concordantly, every one of J_1's closure failures is a product of two half-integer elements (ten of the sixteen products $a_i \cdot a_j$), while all 128 products involving a unit close. The count "eight" is thus not arbitrary: it is exactly the number delivered by the first, weaker sub-condition.

Sources and tracing. This section traces not to Kirmse but to an independent computation. The multiplication is that of Section 2 (Kirmse's table (1)); the lattice J_1 is (3.4)–(3.5) (Kirmse's p. 70). The arithmetic is exact and is reproducible from the scripts `verify_kirmse_1924.py` and `verify_kirmse_1924_forensic.py` (in `python_project/src/`) of the repository [2]; the runs were performed under prompts 100 and 103 of the research record. Item (b)'s witness (5.1) was also checked by hand. The count of 30 doubly-even self-dual binary codes of length 8 is classical; items (c) and (d) re-derive it, and the figures "seven" and "eight," the identity of the eight, and the block in which J_1 fails are all outputs of exact computation, not quotations.

6. Assessment

What Kirmse accomplished in these twenty pages is considerable, and is set out in Sections 2–4: a working octonion algebra; conjugation, norm, and the composition law; the composition identities (2.7) that bear his name; the notions of order and maximal order; the structural Theorem 3.2; an ideal theory; the 240-unit Moufang loop; and, as the paper's announced goal, a derivation of the eight-squares theorem (4.2). Much of the modern arithmetic of the integral octonions is already here, in 1924.

His one substantive error is exactly located. It is not in the algebra, nor in the ideal theory, nor in the final theorem, all of which concern the E_8 lattice as a quadratic form and stand. It is the single claim, (3.3), that there are *eight* maximal orders over J_0. There are seven. And the one maximal order Kirmse wrote down and used, J_1, is itself not multiplicatively closed: of the orders he actually exhibited, the score is none out of one. This is the precise content of what later became known as "Kirmse's mistake." It does not diminish the rest; it is one miscount in a paper that otherwise built a theory.

One matter is deliberately left open here: who detected the error, and in what form the count was corrected. That is the subject of Coxeter's 1946 paper, read from the source in the same manner as Kirmse's and verified in `verify_coxeter_1946.py` (verification note in `paper/reviews/`).

A second matter is largely accounted for. Section 5(d) shows that "eight" is exactly the number of candidate lattices invariant under multiplication by the units, and that those eight are the seven genuine maximal orders together with J_1. Since Kirmse printed no computation, the identification of his "simple finite computation" with this unit-invariance test is a reconstruction, not a quotation, but it reproduces both his count and his choice of J_1 exactly. What is *not* recoverable from his text is the route by which Kirmse himself arrived there. A frankly speculative reconstruction of that route, contributed by J. Köplinger, is set out in Appendix A; it is kept separate precisely because it is speculation, not record.

A small biographical note, since it differs from some secondary accounts: the paper's by-line reads "Johannes Kirmse *in Hannover*," it was presented to the Saxon Academy by G. Herglotz in the session of 5 May 1924, and it is signed "Hannover, am 15. April 1924."

Sources and tracing. The list of accomplishments refers back to Sections 2–4 and through them to Kirmse's §§1–5. The localization of the error refers to (3.3) (Kirmse, p. 70) and to Section 5. The by-line, the presentation by Herglotz, and the place and date of signature are read from [1], pp. 63 and 82. The historical questions are addressed in the verification of H. S. M. Coxeter, *Integral Cayley numbers*, Duke Math. J. **13** (1946), 561–578 (re-derived in `verify_coxeter_1946.py`; companion note 22 May 2026) and in `paper/historical_appendix.tex`.

Appendix A. A speculative note on provenance

This appendix differs in kind from the body. Everything above is either traced to Kirmse's printed text or verified by computation; what follows is neither. It is an openly speculative reconstruction, proposed by Jens Köplinger, of how a human being might have been led to the error of Section 3. It is placed in an appendix, and marked plainly, so that it cannot be mistaken for the record.

Kirmse's paper states its general claim first and only afterwards descends to the particular order J_1. Human inquiry often runs the other way: a concrete instance, found and checked, comes first, and the general statement is distilled from it. One may therefore picture (and it is only a picture) Kirmse proceeding in that order: a worked example in hand; a generalization drawn from it to the form "J_0, half-integer middle elements, $\frac{1}{2}\sum i_k$"; a count of the resulting class; and a representative chosen from it. On this reading the misstep enters at the generalization. Not in the *form*, which is correct (every genuine maximal order has it), but in the leap from a single instance to a complete classification, asserted on a criterion left unspoken. Section 5(d) gives that unspoken criterion a definite shape: closure under the units, a condition necessary for an order but not sufficient, which admits eight lattices where seven are orders. The representative Kirmse then exhibited, J_1, would on this reading have been drawn from a class delimited by a test he never stated, and which does not, in fact, delimit the orders.

What the paper permits, and what it does not, must be said plainly. It does *not* permit the story above to be told as history. Kirmse exhibits exactly one concrete realization, J_1, and J_1 is not an order; if a working first example ever existed, it has left no trace. The order of *exposition* (general, then particular) is not evidence of the order of *discovery*. What is certain is only the logical content of Section 5: closure is a two-part test, and the second part was not carried through. The reconstruction above is consistent with that content, and is offered for no stronger reason than consistency.

Why record a speculation at all. The anatomy of an error (how it is made, how it survives a careful reader, how it is at last detected and corrected) is itself worth study, and a candid hypothesis about the human step, marked unmistakably as hypothesis, can serve that study better than silence. That is the spirit in which this appendix is offered, and the reason it is quarantined from the body.

Sources and tracing. This appendix traces to no source. The speculative reconstruction of a human provenance was contributed by Jens Köplinger; the present author has rendered it in a form consistent with the verified findings of Sections 5–6 and with the rule that speculation be labeled as such. Its only factual anchors (the two-part closure test, the eight unit-invariant lattices, and the block in which J_1 fails) are those of Section 5(d).

References

[1] J. Kirmse, *Über die Darstellbarkeit natürlicher ganzer Zahlen als Summen von acht Quadraten und über ein mit diesem Problem zusammenhängendes nichtkommutatives und nichtassoziatives Zahlensystem*, Berichte über die Verhandlungen der Sächsischen Akademie der Wissenschaften zu Leipzig, Mathematisch-Physische Klasse **76** (1924), 63–82.

[2] J. Köplinger, *An order on the Leech lattice from a $\mathbb{Z}_3$-symmetric triple-octonion product: research repository*, `https://bitbucket.org/jenskoeplinger/leech_alg`, 2026. Verification scripts: `python_project/src/verify_kirmse_1924.py` and `verify_kirmse_1924_forensic.py`; research record: prompts 100, 103, 108.

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